

Random Variable and Probability Distribution

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- ④ Mathematical Expectation and Variance

Random Variable

Introduction

A variable which takes numerical values resulting from random experiments, is called a random variable (RV).

- There is a probability associated with each possible values
- Random variables are denoted by capital letters such as X, Y, Z etc.
- Possible values are denoted by small letters such as x, y, z etc.
- Example:
 - Height of students
 - Number of heads when tossing a coin three times

Types of Random Variable

- **Discrete random variable:** A random variable defined over a discrete sample space
 - Number of students in a class, sample space, $S = \{0, 1, 2, \dots, \infty\}$
 - Number of correct answers in among 50 questions, $S = \{0, 1, 2, \dots, 50\}$
- **Continuous random variable:** A random variable defined over a continuous sample space
 - Monthly income, $S = \{x : 0 \leq x < \infty\}$
 - Monthly profit, $S = \{x : -\infty < x < \infty\}$

Probability Distribution

Probability Distribution

- Probability distribution means the distribution of the probabilities among the different values of a random variable
- For example, when tossing a coin twice, the sample space, $S = \{HH, HT, TH, TT\}$
- Let an RV, X = number of heads in two coin tosses, then X can take the values: 0, 1, 2
- Then the probability of each value of X :

x	$P(X = x)$
0	1/4
1	2/4
2	1/4

Types of Probability Distributions

Depending on the type of variable, distributions are of two types:

- **Discrete probability distribution:** probability distribution of a discrete random variable
- **Continuous probability distribution:** probability distribution of a continuous random variable

Discrete probability distribution

- The probability distribution of a discrete random variable is a table, graph, formula, or other device used to specify all possible values of a discrete random variable along with their respective probabilities
- If we let the discrete probability distribution be represented by the function $p(x)$, then $p(x) = P(X = x)$ is the probability of the discrete random variable X to assume a value x
- $p(x)$ is called a probability mass function (PMF)

Probability Mass Function (PMF)

A probability mass function gives the probability of each of the value of a discrete random variable.

A function, $p(x)$, of a discrete random variable X will be called a PMF, if and only if all the following conditions are satisfied:

① $p(x) \geq 0; \forall x$

② $\sum_x p(x) = 1$

The probability for a value a of the random variable X is written as $p(a)$, i.e., $p(a) = P(X = a)$.

Example: PMF

In page 3, the probability mass function of the variable X representing the number of heads when a coin is tossed twice is given. It is an example of a PMF.

x	$P(X = x)$
0	1/4
1	2/4
2	1/4

Probability Density Function (PDF)

The probability distribution of a continuous random variable is called continuous probability distribution. The probabilities are given by the probability density function (PDF) of that distribution.

A function $f(x)$ of a continuous random variable X is called the PDF of X if and only if the following conditions are satisfied:

- ① $f(x) \geq 0; \forall x$
- ② $\int_x f(x)dx = 1$

Here, $\int_x f(x)dx$ means integration of $f(x)$ over the whole sample space of X . Sometimes, it's also written as $\int_{-\infty}^{\infty} f(x)dx$.

In the continuous case, the probability that $X \in [a, b]$ is found by integrating the PDF as: $P(a \leq X \leq b) = \int_a^b f(x)dx$.

PDF: Example

Suppose X is a continuous random variable whose density function is given by:

$$f(x) = \begin{cases} C(4x - 2x^2) & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

- 1 What is the value of C ?
- 2 Find $P(X > 1)$.

PDF: Example Solution

1

$$\begin{aligned}\int_x f(x)dx &= 1 \\ \Rightarrow \int_0^2 C(4x - 2x^2)dx &= 1 \\ \Rightarrow C[4x^2/2 - 2x^3/3]_0^2 &= 1 \\ \Rightarrow C &= 3/8\end{aligned}$$

2

$$P(X > 1) = \int_1^2 \frac{3}{8}(4x - 2x^2) = 1/2$$

Probability at a Single Point for Continuous Random Variables

For a continuous RV, probability over an interval is:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

At a single point, the interval collapses to zero width, so:

$$P(X = a) = \int_a^a f(x) dx = 0$$

Since $P(X = a) = 0$, adding or removing a boundary point never changes probability:

- $P(X \leq a) = P(X < a)$ and $P(X \geq a) = P(X > a)$
- $P(a \leq X \leq b) = P(a < X < b)$

Cumulative Distribution Function (CDF)

Cumulative Distribution Function (CDF)

The cumulative distribution function (CDF) gives the probability that a random variable X is lower than some value x . Sometimes, it is also called “distribution function”. It is denoted by $F(x)$.

- For discrete random variable, $F(x) = P(X \leq x) = \sum_{k=-\infty}^x p(k)$
- For continuous case, $F(x) = P(X < x) = P(X \leq x) = \int_{-\infty}^x f(x)dx$
- It is a non-decreasing function, the largest value $F(x)$ can take is 1

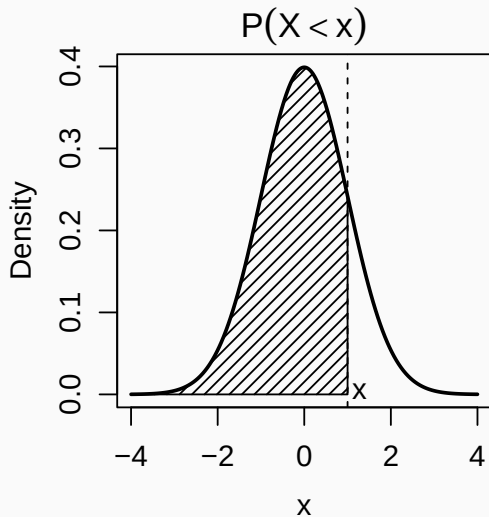
Here, summing/integrating from $-\infty$ means that the lower limit of the sum/integral is the smallest value in the domain of the random variable X .

Graphical Representation of CDF

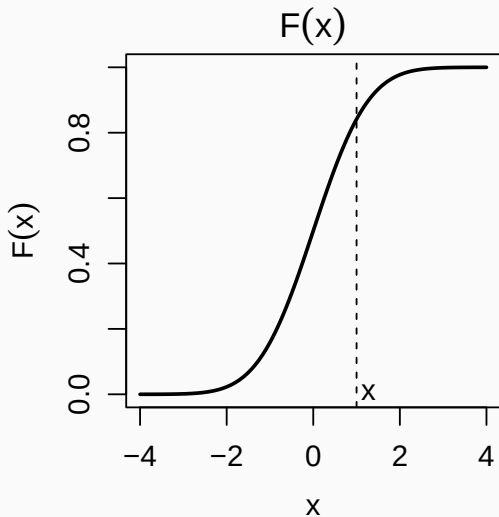
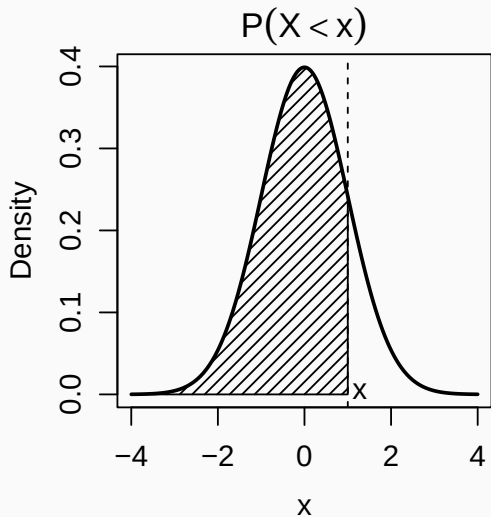
In the plot, X is a continuous random variable, it is “normally” distributed (bell shaped curve).

The shaded region highlights the probability that the random variable X is less than some value x (in the plot, this value is shown to be 1 as an example).

Therefore, the highlighted area is the probability that X is less than 1.



Graphical Representation of CDF (cont.)



CDF: Example 1

In page 3, the probability mass function of the variable X representing the number of heads when a coin is tossed twice is given. Below, its CDF is also given:

x	$P(X = x)$	$P(X \leq x)$
0	1/4	1/4
1	2/4	3/4
2	1/4	1

CDF: Example 2

Suppose X is a continuous random variable whose density function is given by:

$$f(x) = \begin{cases} \frac{3}{8}(4x - 2x^2) & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find the CDF of X :

$$\begin{aligned} F(x) &= \int_0^x f(x)dx \\ &= \int_0^x \frac{3}{8}(4x - 2x^2)dx \\ &= \frac{3}{8}[(4x^2/2 - 2x^3/3)]_0^x = \frac{3}{8}[(2x^2 - 2x^3/3) - 0] = \frac{3}{8}(2x^2 - 2x^3/3) \end{aligned}$$

Finding Probability from CDF

Continuous RV:

- $P(X \leq a) = F(a)$
- $P(X > a) = 1 - F(a)$
- $P(a < X < b) = F(b) - F(a)$

Since $P(X = a) = 0$, the $\leq$ and $<$ signs are interchangeable for continuous RVs.

Discrete RV:

- $P(X \leq a) = F(a)$
- $P(X < a) = F(a) - P(X = a)$
- $P(X > a) = 1 - F(a)$
- $P(a < X \leq b) = F(b) - F(a)$
- $P(a \leq X \leq b) = F(b) - F(a) + P(X = a)$

Example: Finding Probability from CDF of a Continuous RV

Suppose the CDF of a random variable X is

$$F(x) = x^2 \quad \text{for } 0 \leq x \leq 1$$

Find $P(0.2 < X < 0.6)$ and $P(0.2 \leq X \leq 0.6)$.

Using the CDF rule:

$$\begin{aligned} P(0.2 < X < 0.6) &= F(0.6) - F(0.2) = (0.6)^2 - (0.2)^2 \\ &= 0.36 - 0.04 = 0.32. \end{aligned}$$

Since X is continuous, $P(X = a) = 0$.

Therefore, $P(0.2 \leq X \leq 0.6) = 0.32$.

Example: Finding Probability from CDF of a Discrete RV

Suppose the CDF of a random variable X satisfies: $F(2) = 0.35$, $F(5) = 0.80$ and $P(X = 2) = 0.10$. Find $P(2 < X \leq 5)$ and $P(2 \leq X \leq 5)$.

First,

$$\begin{aligned}P(2 < X \leq 5) &= F(5) - F(2) \\&= 0.80 - 0.35 = 0.45.\end{aligned}$$

Next,

$$\begin{aligned}P(2 \leq X \leq 5) &= F(5) - F(2) + P(X = 2) \\&= 0.45 + 0.10 = 0.55.\end{aligned}$$

Mathematical Expectation and Variance

Mathematical Expectation

Expectation of an RV, X is the weighted average of the possible values of X . The weight of each value of X is its corresponding probability.

- When X is discrete: $E(X) = \sum_x x \cdot p(x)$
- When continuous: $E(X) = \int_x x \cdot f(x) dx$
- The sum/integral is over all values of X
- It is often denoted by μ , that is: $E(X) = \mu$
- It is also known as expected value or population mean

Variance of a Random Variable

The variance of an RV, X measures the dispersion in the values of X . It is denoted by $\text{Var}(X)$ or $V(X)$.

$$\begin{aligned} V(X) &= E[(X - E(X))^2] \\ &= E[X^2 - 2X \cdot E(X) + [E(X)]^2] \\ &= E(X^2) - 2E(X) \cdot E(X) + [E(X)]^2 \\ &= E(X^2) - 2[E(X)]^2 + [E(X)]^2 \\ &= E(X^2) - [E(X)]^2 \end{aligned}$$

Here, $E(X^2)$ is given by $\sum_x x^2 \cdot p(x)$ or $\int_x x^2 \cdot f(x) dx$ for discrete and continuous case respectively.

Expectation and Variance: Example 1

In page 3, the PMF of the random variable X representing the number of heads in two throws of a fair coin is given.

X can take the values 0, 1 and 2 with probabilities $1/4$, $1/2$ and $1/4$ respectively.

The expected value of X is:

$$E(X) = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 0 + \frac{1}{2} + \frac{1}{2} = 1$$

The variance of X is:

$$\begin{aligned} V(X) &= E(X^2) - [E(X)]^2 \\ &= \left[0^2 \cdot \frac{1}{4} + 1^2 \cdot \frac{1}{2} + 2^2 \cdot \frac{1}{4} \right] - 1^2 \\ &= \left[0 + \frac{1}{2} + 1 \right] - 1 = \frac{1}{2} \end{aligned}$$

Expectation and Variance: Example 2

Find the expected value and the variance of the RV X whose PDF is given in page 16.

$$\begin{aligned} E(X) &= \int_0^2 x \cdot \frac{3}{8}(4x - 2x^2) dx \\ &= \frac{3}{8} \int_0^2 (4x^2 - 2x^3) dx \\ &= \frac{3}{8} [4x^3/3 - 2x^4/4]_0^2 \\ &= \frac{3}{8} [4 \times 2^3/3 - 2 \times 2^4/4 - 0] = 1 \end{aligned}$$

Finding the variance is left as an exercise.

Properties of Expectation and Variance

Let a , b and C be some constants, and X and Y be two independent random variables.

$$\textcircled{1} \ E(c) = c$$

$$\textcircled{2} \ E(cX) = c E(X)$$

$$\textcircled{3} \ E(c + X) = c + E(X)$$

$$\textcircled{4} \ E(aX + bY) = a E(X) + b E(Y)$$

$$\textcircled{1} \ V(c) = 0$$

$$\textcircled{2} \ V(cX) = c^2 V(X)$$

$$\textcircled{3} \ V(c + X) = V(X)$$

$$\textcircled{4} \ V(aX + bY) = a^2 V(X) + b^2 V(Y)$$

If X and Y are dependent:

$$V(aX + bY) = a^2 V(X) + b^2 V(Y) + 2ab \text{CoV}(X, Y).$$

Here, $\text{CoV}(X, Y)$ is the covariance of X and Y . More on covariance in a future lecture.

Properties of Expectation and Variance: Example 1

Suppose X is a random variable with $E(X) = 5$ and $V(X) = 2$.

- $E(2X) = 2E(X) = 2 \times 5 = 10$
- $V(2X) = 2^2 V(X) = 4 \times 2 = 8$
- $E(2X - 3) = 2E(X) - 3 = 2 \times 5 - 3 = 10 - 3 = 7$
- $V(2X - 3) = 2^2 V(X) = 4 \times 2 = 8$

Thank you.
